Overview of PertOmni

a

Drug-perturbation dataset

Chemical/Gene information…

LLM

Text Encoder

EXP Encoder

Gene-perturbation dataset

Cell painting…

FM

Visual Encoder

T_1 T_2 T_3 $T_{...}$ T_n

E_1 E_2 E_3 $E_{...}$ E_n

E_1' E_2' E_m'

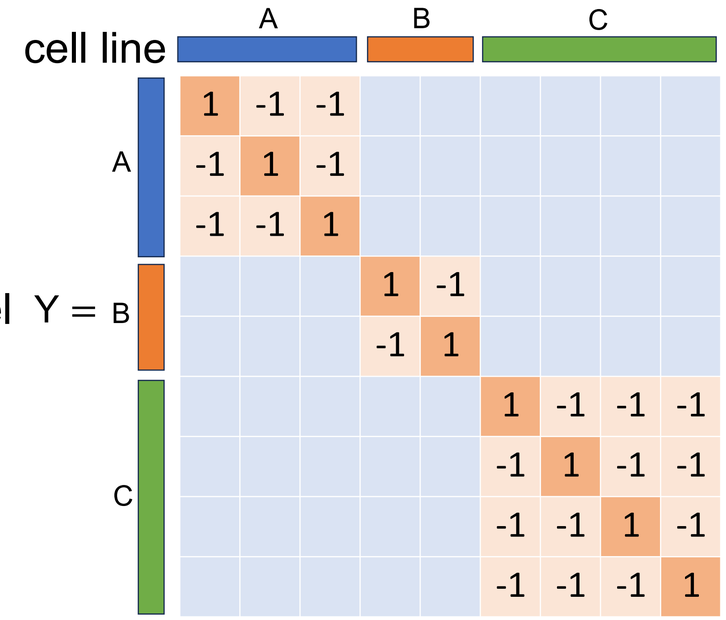
E_1T_1 E_2T_2 E_3T_3 E_nT_n

$E_1'M_1$ $E_2'M_2$ $E_m'M_m$

M_1 M_2 M_3 $M_{...}$ M_m

L_{mCLIP}

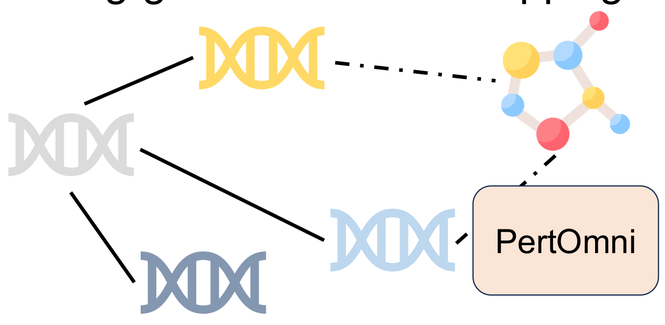
b



$$L_{mCLIP} = -\frac{1}{n}\sum_{i,j} \log\sigma(Y_{ij} \cdot \langle E_i, T_j \rangle / t) - \frac{1}{m}\sum_{i',j'} \log\sigma(Y'_{i',j'} \cdot \langle E'_{i'}, M_{j'} \rangle / t)$$

c

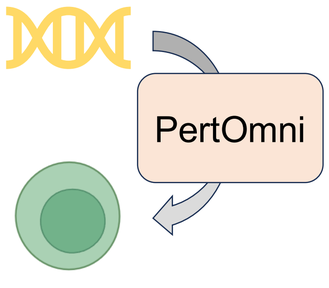
Drug-gene interaction mapping



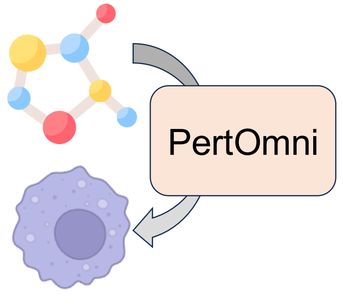
Perturbation prediction



Genetic perturbation retrieval



Chemical perturbation retrieval



Cell Painting image retrieval

